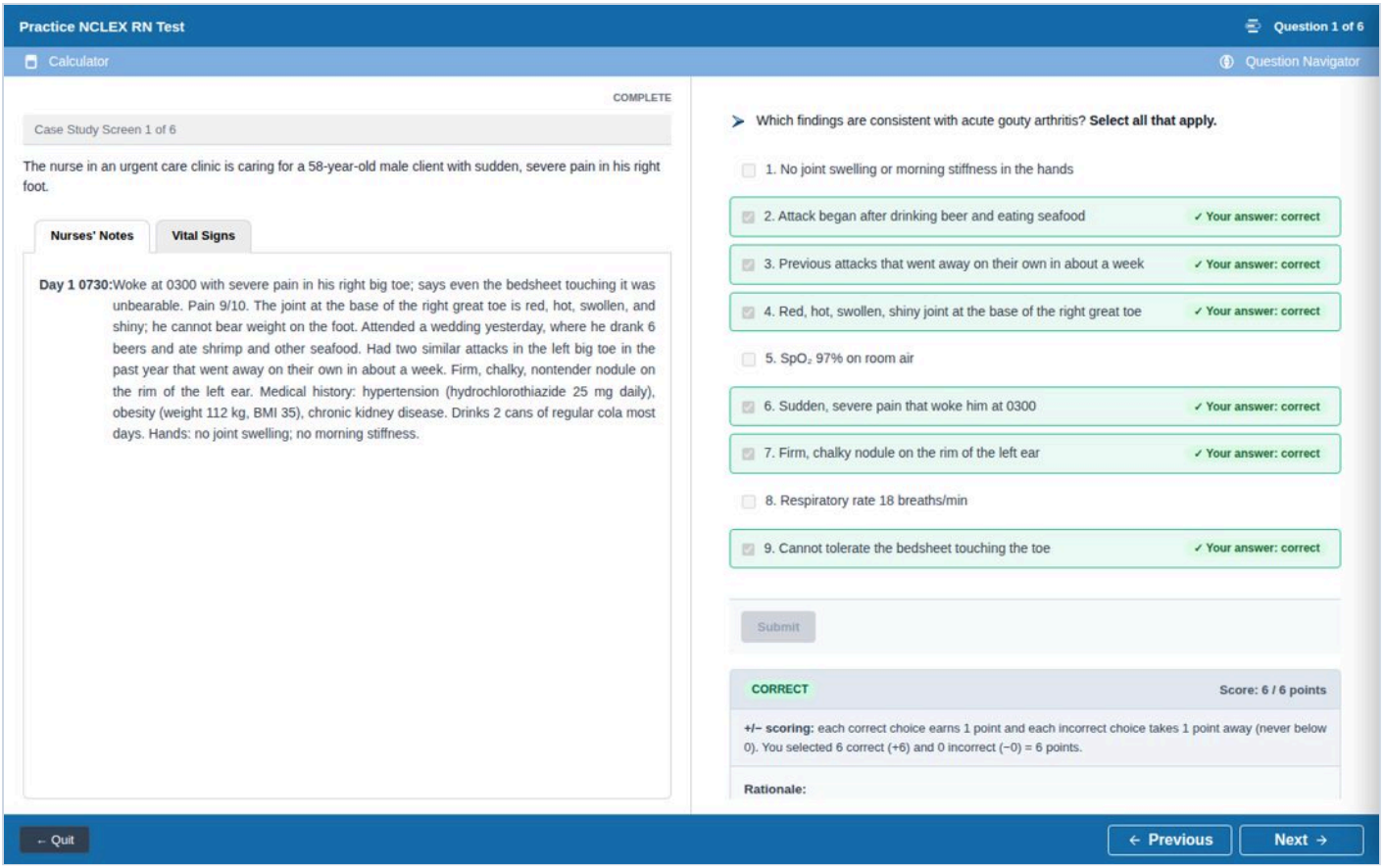
NURS 1017 Unit 6 Case Studies 8 and 9 (review copy)

For each screen: the question with the answer key filled in and submitted, then the full rationale as students see it after Submit. All 12 screens score full marks with their answer keys and both cases pass the readiness check.

- Case Study 8: Gout
- Case Study 9: Ankylosing spondylitis

Case Study 8 (Gout), screen 1 of 6: Recognize cues



Case Study 8, screen 1: rationale

CORRECT

Score: 6 / 6 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

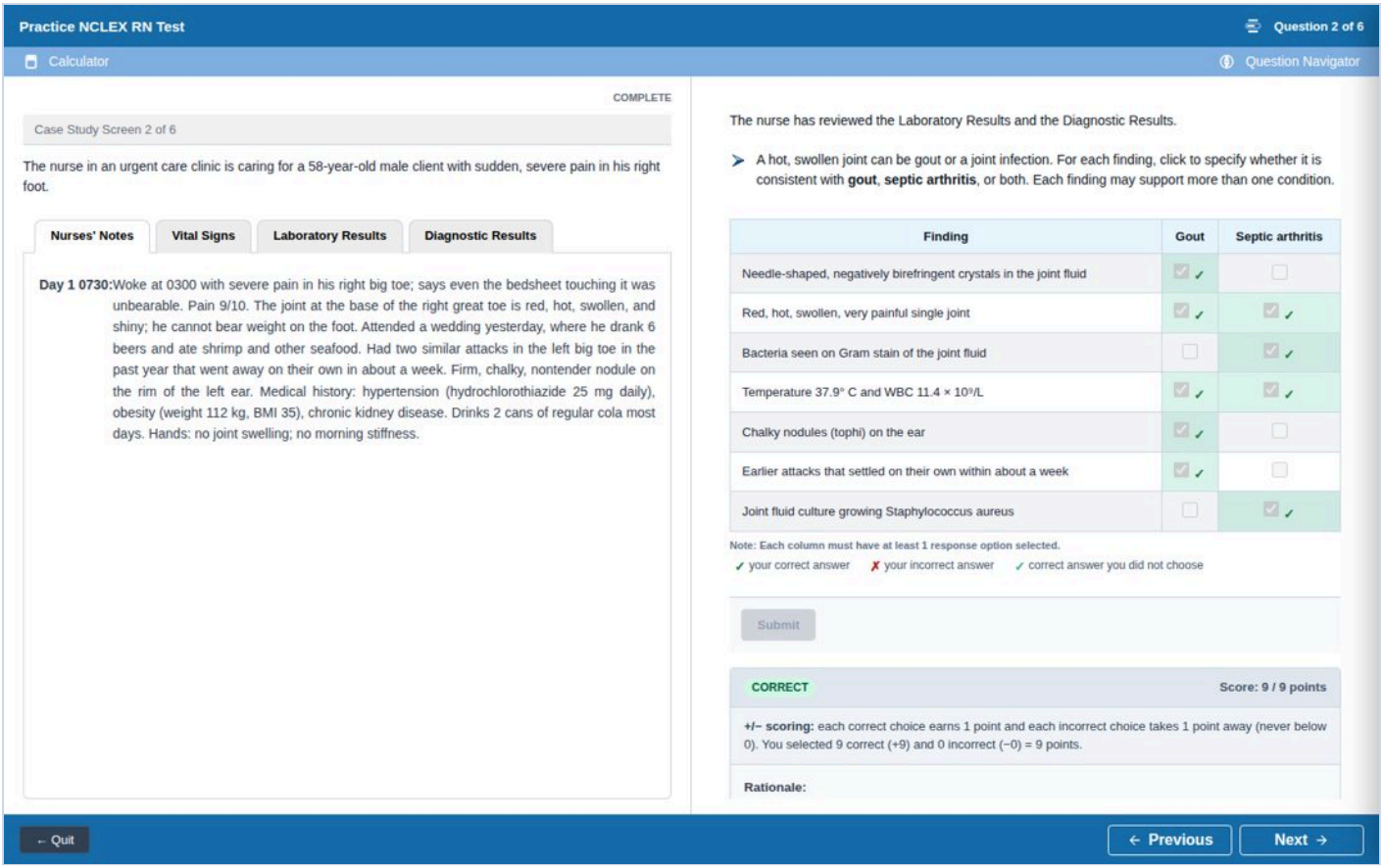
Typical of acute gout: attacks start suddenly, often at night, and reach maximal pain within 12 to 24 hours. The first metatarsophalangeal joint of the big toe is the classic site (**podagra**). The joint is red, hot, swollen and shiny and so tender that even a bedsheet hurts. Untreated attacks usually settle in 1 to 2 weeks, and the client is symptom-free between attacks.

Triggers: alcohol (especially beer, which is rich in purines) and purine-rich foods such as shellfish and organ meats raise uric acid; dehydration, fasting and surgery can also trigger attacks.

Tophi: firm, chalky deposits of urate crystals under the skin (ear rim, elbows, fingers) show long-standing high urate levels (chronic tophaceous gout). A low-grade fever can accompany an acute attack.

Not cues: the absence of hand involvement and morning stiffness, and his normal breathing and oxygen saturation, do not point toward gout.

Case Study 8 (Gout), screen 2 of 6: Analyze cues



Case Study 8, screen 2: rationale

CORRECT **Score: 9 / 9 points**

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 9 correct (+9) and 0 incorrect (−0) = 9 points.

Rationale:

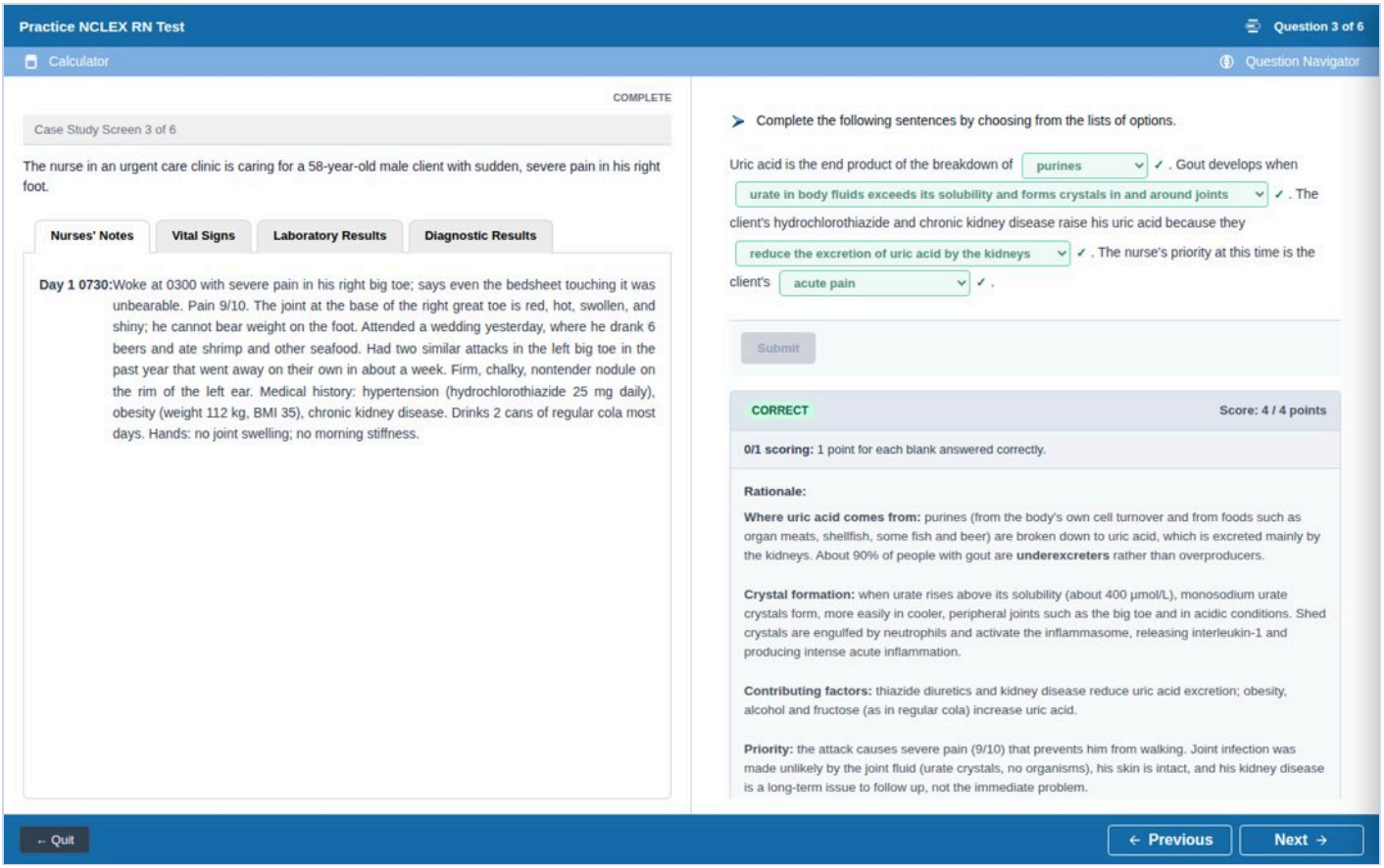
Both: gout and septic arthritis can both produce a single red, hot, swollen, very painful joint with fever, a raised WBC count and raised CRP. Because an infected joint can be destroyed within days, joint fluid is aspirated to tell them apart; the two can also occur together.

Gout: the diagnostic finding is **monosodium urate crystals**, needle-shaped and negatively birefringent under polarized light, inside neutrophils in the joint fluid. Tophi and earlier attacks that resolved on their own support gout. This client's joint fluid showed urate crystals and no organisms on Gram stain; the culture is pending.

Septic arthritis: bacteria on Gram stain or a positive culture (most often *Staphylococcus aureus*) confirm infection. Septic arthritis does not settle on its own and needs urgent treatment.

Note on uric acid: his serum uric acid is high (528 µmol/L), but it can be normal during an acute attack, so a normal level does not rule out gout; the crystals do.

Case Study 8 (Gout), screen 3 of 6: Prioritize hypotheses



Case Study 8, screen 3: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

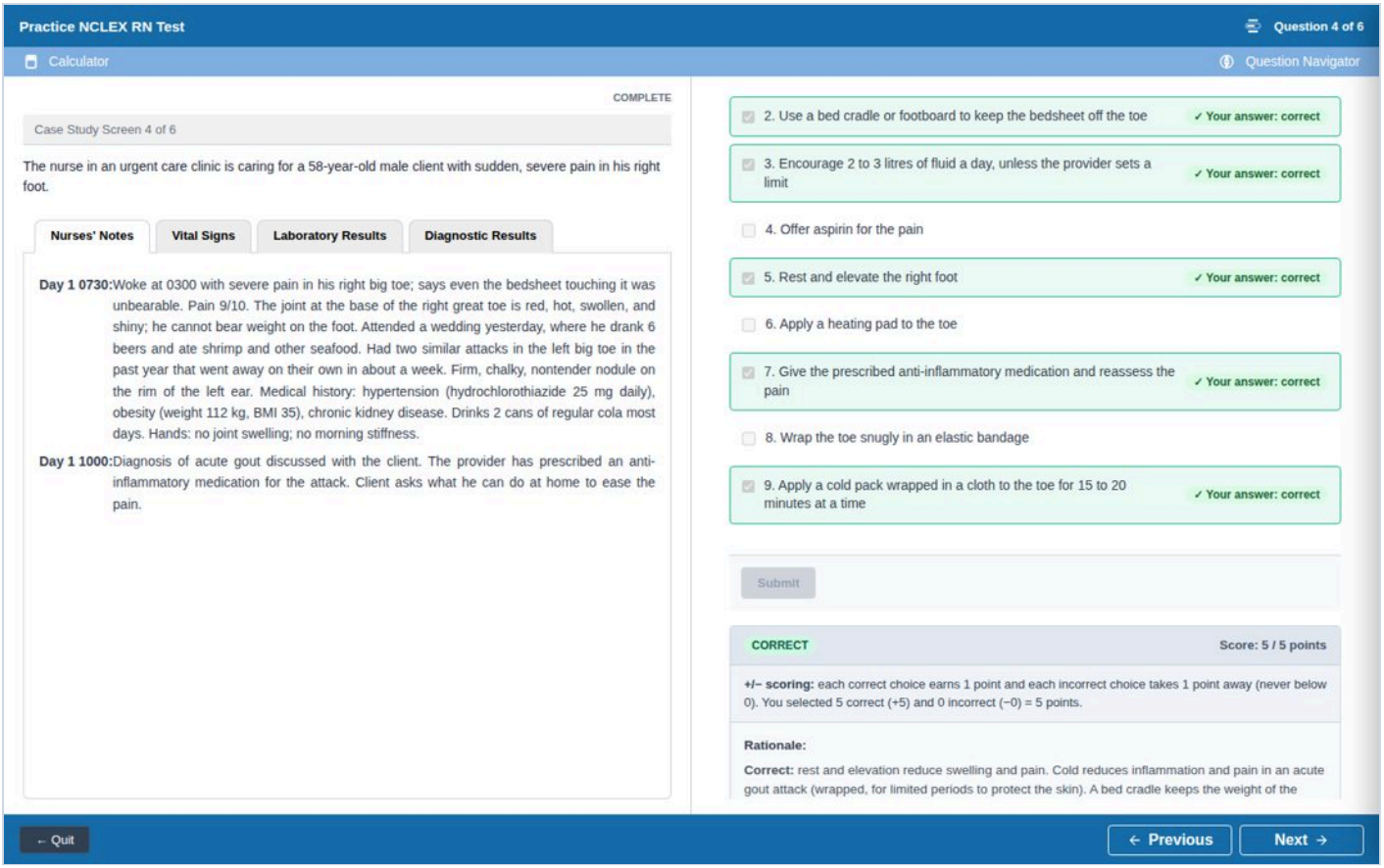
Where uric acid comes from: purines (from the body's own cell turnover and from foods such as organ meats, shellfish, some fish and beer) are broken down to uric acid, which is excreted mainly by the kidneys. About 90% of people with gout are **underexcreters** rather than overproducers.

Crystal formation: when urate rises above its solubility (about 400 µmol/L), monosodium urate crystals form, more easily in cooler, peripheral joints such as the big toe and in acidic conditions. Shed crystals are engulfed by neutrophils and activate the inflammasome, releasing interleukin-1 and producing intense acute inflammation.

Contributing factors: thiazide diuretics and kidney disease reduce uric acid excretion; obesity, alcohol and fructose (as in regular cola) increase uric acid.

Priority: the attack causes severe pain (9/10) that prevents him from walking. Joint infection was made unlikely by the joint fluid (urate crystals, no organisms), his skin is intact, and his kidney disease is a long-term issue to follow up, not the immediate problem.

Case Study 8 (Gout), screen 4 of 6: Generate solutions



Case Study 8, screen 4: rationale

CORRECT

Score: 5 / 5 points

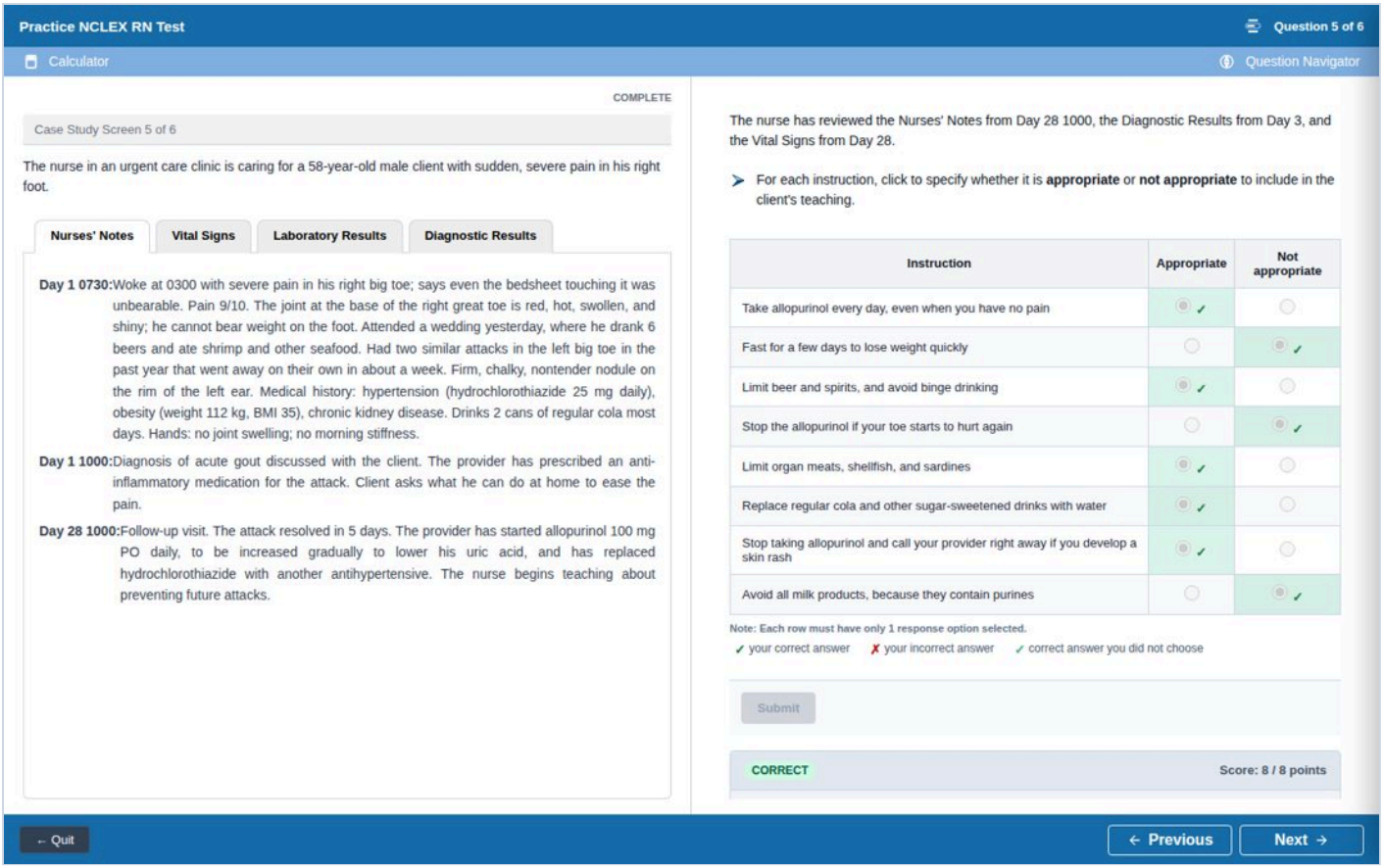
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Correct: rest and elevation reduce swelling and pain. Cold reduces inflammation and pain in an acute gout attack (wrapped, for limited periods to protect the skin). A bed cradle keeps the weight of the sheets off the exquisitely tender joint. Taking the prescribed medication early in the attack shortens it; the nurse reassesses pain afterward. Fluids help the kidneys excrete uric acid and reduce the risk of urate kidney stones; because he has chronic kidney disease, the nurse checks whether the provider has set a fluid limit.

Incorrect: heat increases the inflammation of an acute attack. Massage or pressure on the joint is extremely painful and does not remove crystals. Aspirin (salicylate) can raise uric acid levels by competing with its excretion in the kidney, so it is avoided for gout pain. A snug bandage puts pressure on the swollen, tender joint.

Case Study 8 (Gout), screen 5 of 6: Take action



Case Study 8, screen 5: rationale

CORRECT

Score: 8 / 8 points

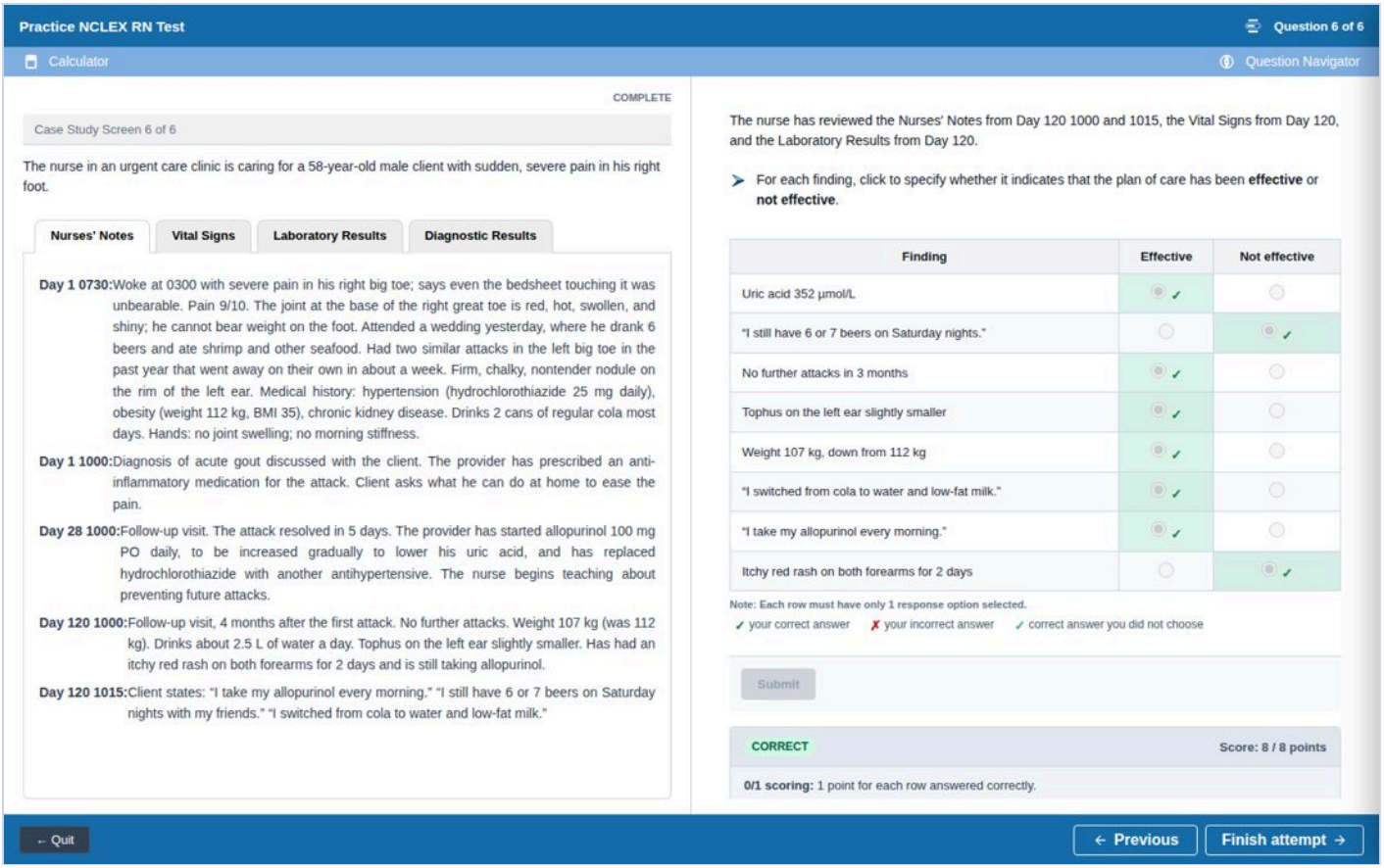
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Appropriate: allopurinol lowers uric acid production and must be taken every day to prevent attacks and dissolve tophi; it does not relieve an attack. Beer and spirits raise uric acid; organ meats, shellfish and some oily fish are high in purines; fructose in sugar-sweetened drinks increases uric acid production. A rash on allopurinol can be the first sign of a rare but life-threatening hypersensitivity reaction (including Stevens-Johnson syndrome), so the drug is stopped and the rash reported at once.

Not appropriate: fasting and crash diets raise uric acid (ketones compete with its excretion) and can trigger attacks; weight loss should be gradual. Attacks can still happen in the first months of urate-lowering therapy as crystals dissolve; stopping allopurinol during an attack makes urate levels swing and prolongs the problem, so the client calls the provider instead. Low-fat dairy products are low in purines and are associated with lower uric acid levels, so they are encouraged.

Case Study 8 (Gout), screen 6 of 6: Evaluate outcomes



Case Study 8, screen 6: rationale

CORRECT

Score: 8 / 8 points

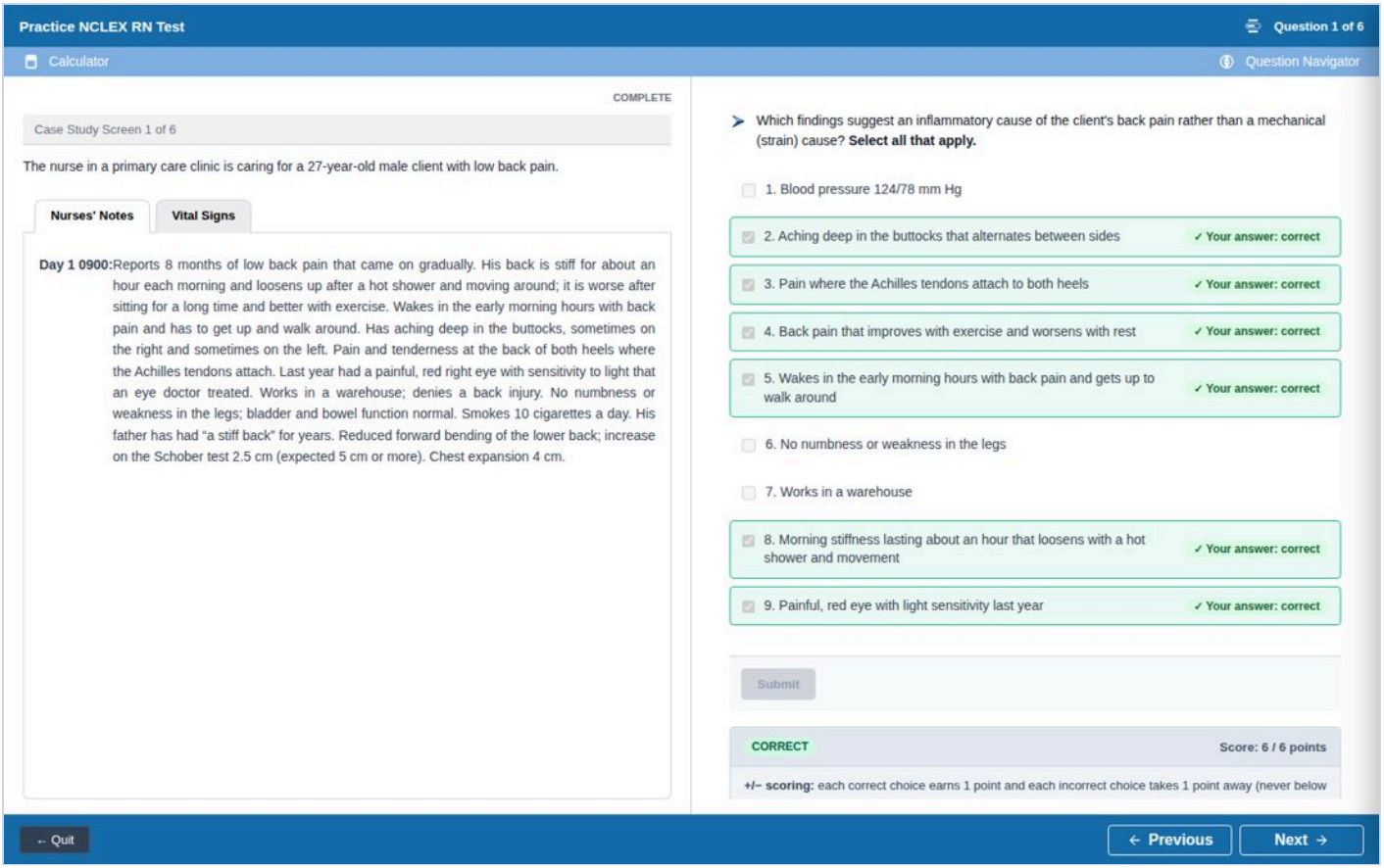
0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: a uric acid below 360 µmol/L is the usual treatment target, low enough for crystals to dissolve; the shrinking tophus shows this is happening. He has had no further attacks, is losing weight gradually, takes allopurinol daily, and has replaced sugary drinks.

Not effective: binge drinking of beer raises uric acid and dehydrates him, and is a common trigger of attacks. The nurse reinforces limiting alcohol and explores support for cutting down. A new rash while taking allopurinol must not be ignored: as taught, the drug is held and the provider is notified the same day, because a rash can be the first sign of a serious hypersensitivity reaction. His kidney function remains reduced and continues to be monitored.

Case Study 9 (Ankylosing spondylitis), screen 1 of 6: Recognize cues



Case Study 9, screen 1: rationale

CORRECT

Score: 6 / 6 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Inflammatory back pain typically starts before age 40, comes on gradually, lasts more than 3 months, causes morning stiffness of 30 minutes or more, improves with exercise but not with rest, and wakes the person in the second half of the night. Mechanical back pain is the opposite: it worsens with activity and improves with rest.

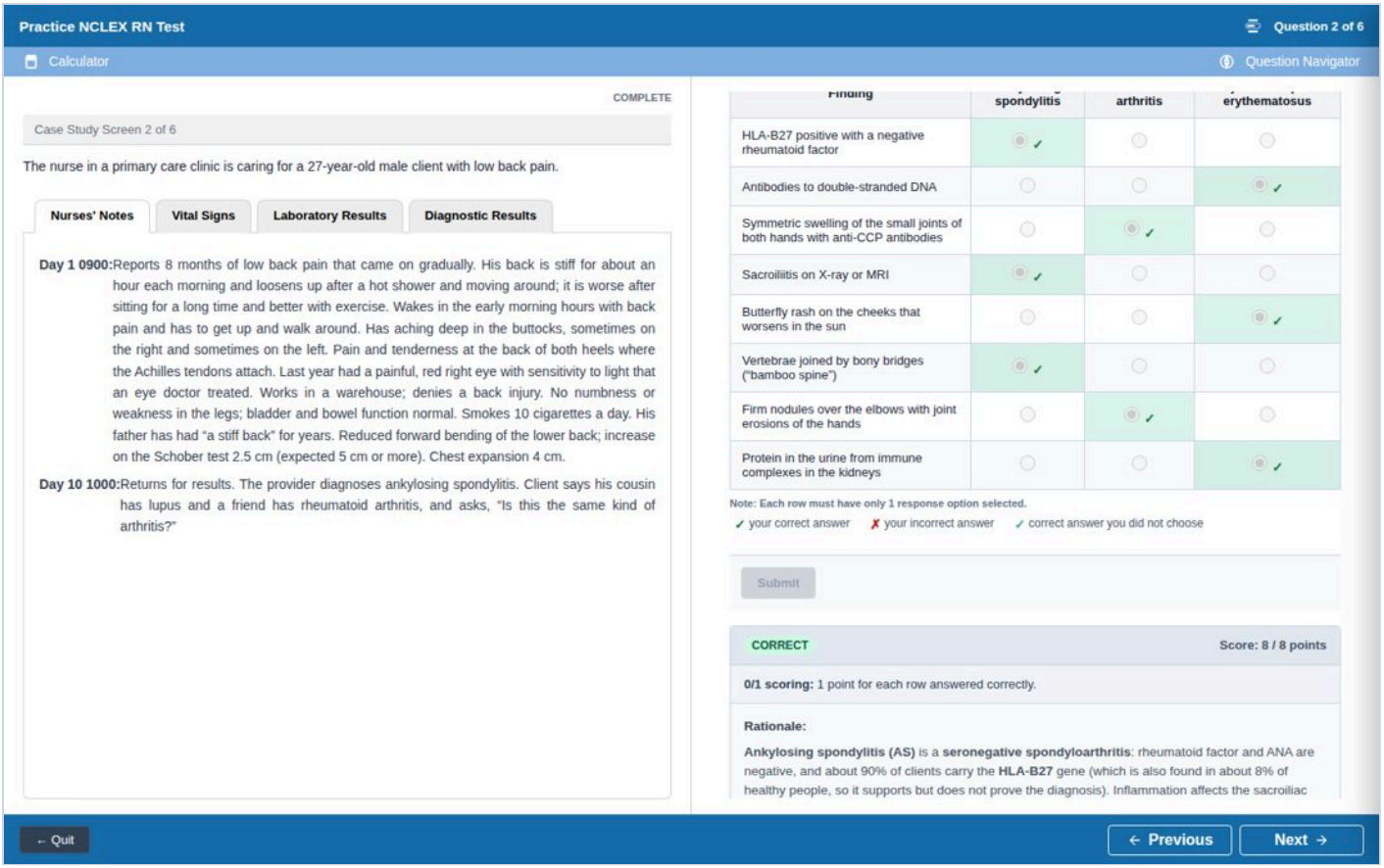
Alternating buttock pain points to inflammation of the sacroiliac joints (sacroiliitis), the hallmark of ankylosing spondylitis (AS).

Heel pain at the Achilles insertion is enthesitis, inflammation where tendons and ligaments attach to bone, a key feature of the spondyloarthritis family.

Acute anterior uveitis (painful red eye with light sensitivity) is the most common extra-articular feature of AS.

Not cues: warehouse work could suggest a mechanical cause, and normal leg strength and sensation with normal bladder and bowel function show no nerve compression; neither points to inflammation. His blood pressure is normal.

Case Study 9 (Ankylosing spondylitis), screen 2 of 6: Analyze cues



Case Study 9, screen 2: rationale

CORRECT

Score: 8 / 8 points

0/1 scoring: 1 point for each row answered correctly.

Rationale:

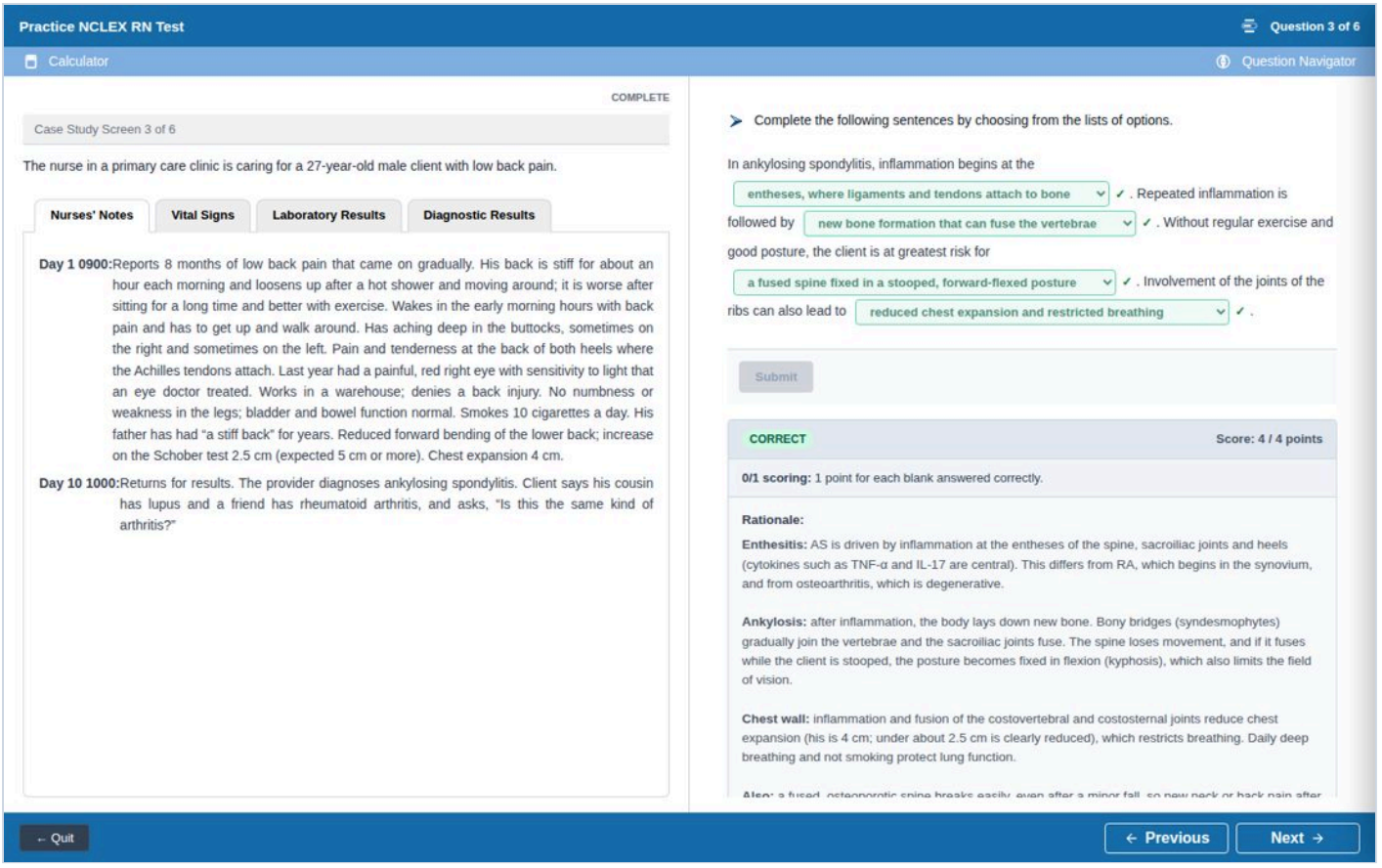
Ankylosing spondylitis (AS) is a **seronegative spondyloarthritis**: rheumatoid factor and ANA are negative, and about 90% of clients carry the **HLA-B27** gene (which is also found in about 8% of healthy people, so it supports but does not prove the diagnosis). Inflammation affects the sacroiliac joints and spine; over years, new bone bridges the vertebrae (syndesmophytes), giving the "bamboo spine" on X-ray. It usually begins in young men.

Rheumatoid arthritis (RA) is a symmetric inflammatory polyarthritis of small joints (hands, wrists, feet) with rheumatoid factor and anti-CCP antibodies, erosions, and rheumatoid nodules.

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease, most common in young women, with ANA and anti-double-stranded DNA antibodies. Immune complexes deposit in skin (malar rash, photosensitivity), joints (non-erosive arthritis), kidneys (lupus nephritis, proteinuria), blood cells and serous membranes.

So all three are inflammatory arthritides, but they differ in the joints involved, the antibodies, and the organs affected.

Case Study 9 (Ankylosing spondylitis), screen 3 of 6: Prioritize hypotheses



Case Study 9, screen 3: rationale

CORRECT **Score: 4 / 4 points**

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

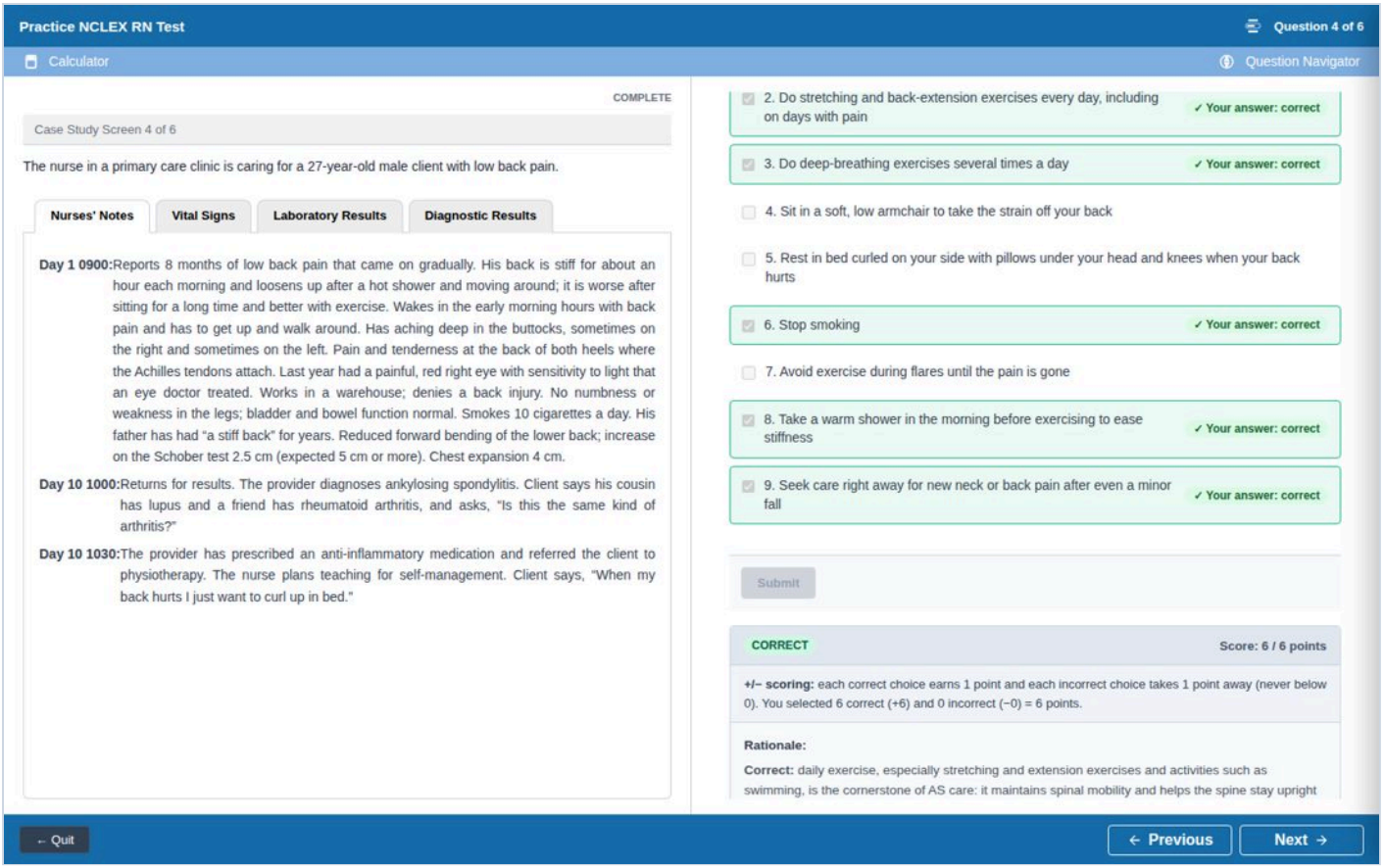
Enthesitis: AS is driven by inflammation at the entheses of the spine, sacroiliac joints and heels (cytokines such as TNF-α and IL-17 are central). This differs from RA, which begins in the synovium, and from osteoarthritis, which is degenerative.

Ankylosis: after inflammation, the body lays down new bone. Bony bridges (syndesmophytes) gradually join the vertebrae and the sacroiliac joints fuse. The spine loses movement, and if it fuses while the client is stooped, the posture becomes fixed in flexion (kyphosis), which also limits the field of vision.

Chest wall: inflammation and fusion of the costovertebral and costosternal joints reduce chest expansion (his is 4 cm; under about 2.5 cm is clearly reduced), which restricts breathing. Daily deep breathing and not smoking protect lung function.

Also: a fused, osteoporotic spine breaks easily, even after a minor fall, so new neck or back pain after trauma needs urgent assessment.

Case Study 9 (Ankylosing spondylitis), screen 4 of 6: Generate solutions



Case Study 9, screen 4: rationale

CORRECT

Score: 6 / 6 points

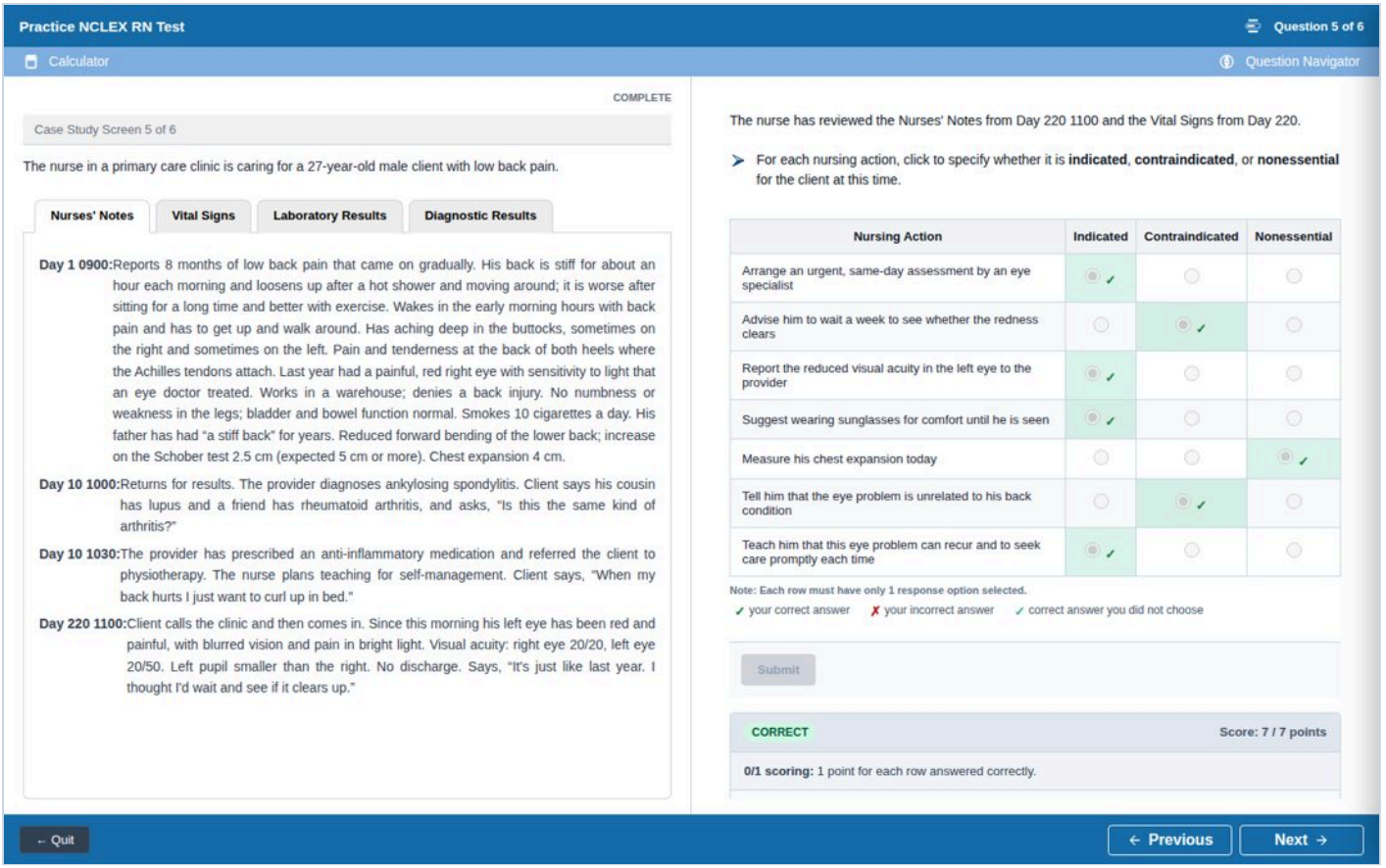
+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Correct: daily exercise, especially stretching and extension exercises and activities such as swimming, is the cornerstone of AS care: it maintains spinal mobility and helps the spine stay upright if it fuses. Sleeping flat on a firm surface with little or no pillow (lying prone for periods each day) prevents flexion deformity. Deep breathing maintains chest expansion. Smoking worsens AS progression and lung function. Heat eases morning stiffness before activity. A fused spine fractures easily, so new pain after a fall must be assessed urgently.

Incorrect: curling up with pillows under the head and knees and sitting in soft, low chairs encourage a flexed posture, and rest worsens inflammatory stiffness. In AS, activity relieves the pain, so exercise continues during flares at a gentler level.

Case Study 9 (Ankylosing spondylitis), screen 5 of 6: Take action



Case Study 9, screen 5: rationale

CORRECT

Score: 7 / 7 points

0/1 scoring: 1 point for each row answered correctly.

Rationale:

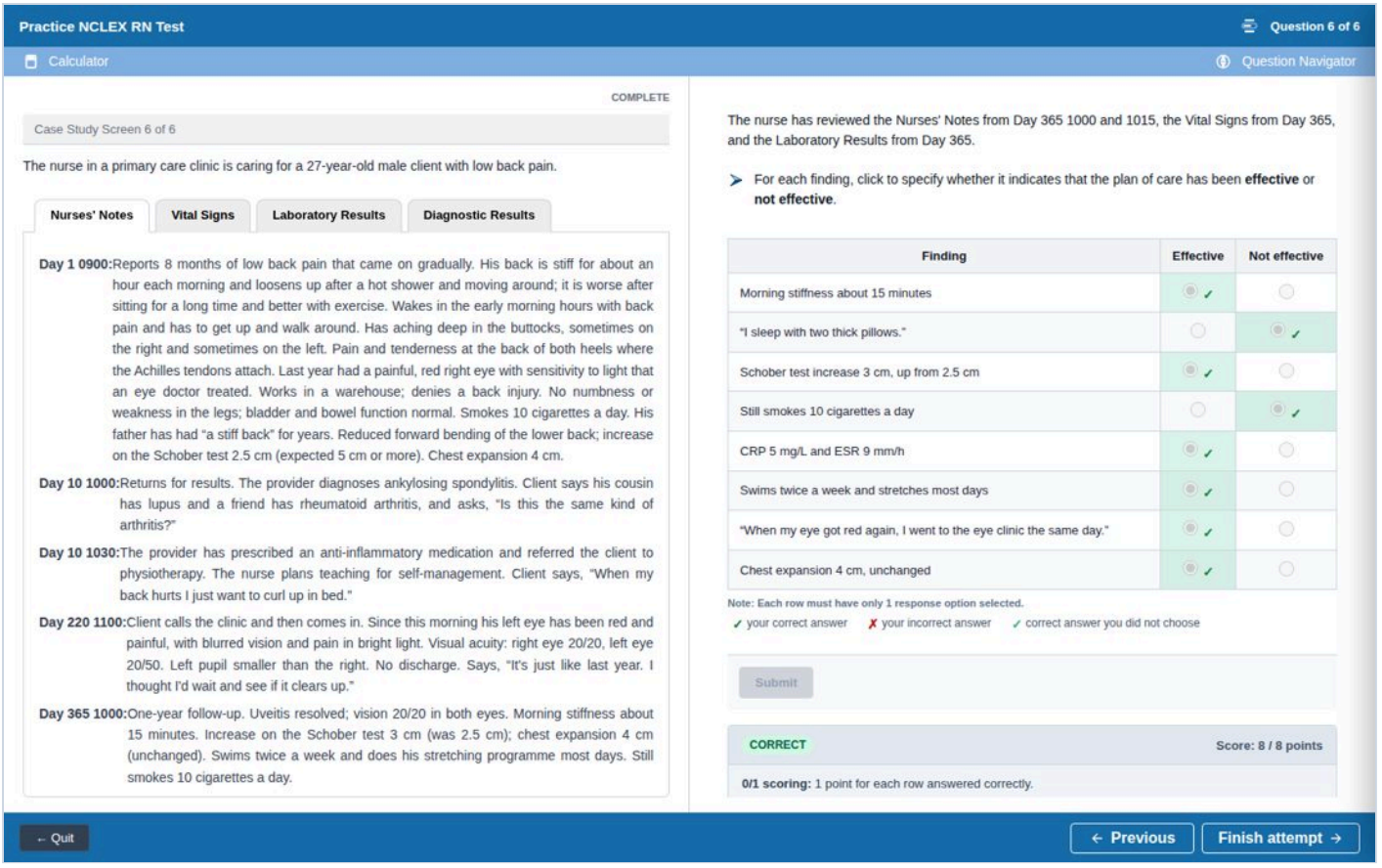
These are signs of **acute anterior uveitis** (iritis): a painful red eye, light sensitivity, blurred vision and a small pupil. It affects up to about a third of people with AS and is often recurrent and on one side.

Indicated: uveitis needs urgent assessment and treatment by an eye specialist to prevent complications such as adhesions of the iris, glaucoma and vision loss; the drop in visual acuity is reported. Sunglasses ease light sensitivity while he waits. Because attacks recur, he is taught to seek care promptly each time.

Contraindicated: waiting risks permanent damage to the eye. Saying the eye problem is unrelated is incorrect: uveitis is a recognized extra-articular feature of AS and shares the HLA-B27 link.

Nonessential: chest expansion is part of routine follow-up, not of this urgent visit.

Case Study 9 (Ankylosing spondylitis), screen 6 of 6: Evaluate outcomes



Case Study 9, screen 6: rationale

CORRECT **Score: 8 / 8 points**

0/1 scoring: 1 point for each row answered correctly.

Rationale:

Effective: shorter morning stiffness and normal inflammatory markers show that inflammation is controlled; spinal mobility has improved slightly and chest expansion has not declined; he exercises regularly; and he recognized a recurrence of uveitis and sought care the same day, as taught.

Not effective: thick pillows hold the neck in flexion and promote a fixed, stooped posture; the nurse reteaches sleeping flat with a thin pillow. Continued smoking speeds spinal fusion and harms lung function in a client whose chest expansion is already limited; the nurse offers cessation support.